

Residential virtual power plant with photovoltaic output forecasting and demand response

Shichang Cui^{1,2} | Yan-Wu Wang^{1,2,3}  | Xiangning Lin⁴ | Jiang-Wen Xiao^{1,2}

¹School of Automation, Huazhong University of Science and Technology, Wuhan, China

²Key Laboratory of Image Processing and Intelligent Control, Ministry of Education, Huazhong University of Science and Technology, Wuhan, China

³Hubei Provincial Collaborative Innovation Center for New Energy Microgrid, China Three Gorges University, Yichang, China

⁴State Key Laboratory of Advanced Electromagnetic Engineering and Technology, Huazhong University of Science and Technology, Wuhan, China

Correspondence

Yan-Wu Wang, School of Automation, Huazhong University of Science and Technology, Wuhan, China.
Email: wangyw@hust.edu.cn

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Abstract

Decreasing conventional power supply is promoting the development of the distributed renewable energy sources, such as solar power and wind power. Recently rooftop photovoltaic has been widely applied, and accordingly efficient energy management is getting increasingly important for fully use of renewable energy and the peak shaving of the main grid. This paper investigates the residential energy management as a small-scale virtual power plant (VPP) connected to the main grid includes distributed energy resources, energy storages and residential loads. The self-organizing map (SOM) and the radical basis function (RBF) networks are adopted to classify the weather types and predict hourly photovoltaic output precisely. In a time-of-use electricity market, price-based demand response is applied to adjust the demand. The residential VPP has two goals: maximum profit by selling surplus power to grid and minimum power purchased from grid. The two goals are integrated as an optimization object by introducing a weight parameter. The algorithm combining receding horizon optimization and linear programming is proposed to solve the optimization problem in residential VPP. Numerical simulation tests can help to find the most suitable value of the weight parameter. Different scenarios are simulated and discussed to demonstrate the performance of the VPP and the proposed algorithm.

KEYWORDS

demand response, energy management, neural network, photovoltaic, receding horizon optimization, virtual power plant

1 | INTRODUCTION

In recently years, ever-increasing energy demand and pressing environmental issues have encouraged the development of renewable energy resources, e.g. solar and wind power. Besides increasing large-scale renewable energy penetration to power systems, small-scale residential rooftop photovoltaic systems have also grown rapidly

due to the advances in electricity generation and storage technologies coupled with declines in cost, the planned roll-out of smart metering and favourable regulation [1-3]. The characteristics of randomness and intermittency and the feed-in tariff bring challenge and opportunity for residential energy management. This paper investigates residential energy management from the perspective of virtual power plant (VPP) which has been widely studied [4-7] in

recent years as an efficient way of aggregating uncertain renewable energy resource, load, demand response and energy storage. These elements are interconnected within a small-size grid-connected electricity energy system. The small energy system is equipped with Smart Grid technology [8], which provides possibility for the residential VPP to make efficient and profitable aggregated energy management scheme [3,9,10].

To overcome the negative effect of uncertain renewable energy output, residential VPP has to adopt precise forecasting methods for rooftop solar power production. The forecasting probabilistic methods includes Monte Carlo simulations, Point Estimate Method, and scenario-based decision making [11], and the operational complexity, the requirement of a large amount of historical data and multiple scenarios make these methods unsuitable for short-term rooftop PV forecasting. The statistical methods such as autoregressive integrated moving average (ARIMA) are not suitable for PV prediction with low spatial resolution [12,13]. With the development of smart algorithms, advanced forecasting techniques, e.g. fuzzy logic methodology, artificial neural network, support vector machines have been commonly adopted [14–17]. For a residential VPP, making energy management scheme requires accurate 24-hour online forecasting, and [16–18] have verified the accuracy and efficiency of artificial neural networks in this application. [17,19] provide more accurate forecasting models based on the weather type classification. Inspired by these work, an accurate hourly PV output forecasting model of multiple radical basis function neural networks with self-organizing map classifying different weather types is presented.

Demand response can lead to peak-valley shifting and reduce the power consumption costs, and can be conducive to promote VPP power generation and make appropriate power schedule. Some literatures discussed how demand response participates in the VPP power scheduling. Demand response, divided into price-based demand response (PBDR) and incentive-based demand response (IBDR), can optimize customers behavior to respond to power optimization scheduling [20]. The PBDR is applied for customers to transfer the demand loads from periods of high price to favourable periods of low prices in an economical way, while the IBDR is provided by DR providers and large customers to participate in energy scheduling of distribution network. Therefore, we only apply the PBDR for residential energy management with one single energy system. [21] studies how to adjust energy scheduling to maximize VPP operation profit by allocating power output. [22] analyses the economic impacts of small-scale generating and storage system under demand response strategies for households. Price-based demand response is applied in our work at a time-of-use (TOU) electricity market, and its

positive effect on residential VPP power consumption and profit will be discussed.

Residential VPP has two goals in our work. One is to maximize the profit by selling surplus power energy from rooftop PV system and storage to main grid, while the other one is to minimize energy interactive with the main grid. By setting a weight parameter, single optimization goal can be obtained. Considering the forecasting error of the hourly PV output power and the difference between the estimated and actual storage state, the algorithm combining receding horizon optimization and linear programming is presented to provide hourly power scheduling according to latest system information. Therefore, the main contributions of this paper are summarized as follows:

1. This paper investigates the residential energy system equipped with rooftop photovoltaic and energy storage as a small-scale virtual power plant. The proposed optimization algorithm combining receding horizon optimization and linear programming provides the optimal hourly-updated optimal power scheduling for residential VPP to achieve VPP goals.
2. Artificial neural network, as a precise way of forecasting PV output power, is firstly applied to VPP to handle the uncertainty of renewable energy by providing hourly power output reference. Moreover, demand response is also adopted in the VPP to optimize the power consumption, which can reduce the power consumption costs and achieve peak valley shifting.

The rest of the paper is organized as follows. Section II analyses 24-hour PV output power forecasting method, price-based demand response and the receding horizon optimization. Section III provides the VPP model in detail. Section IV presents simulation details and optimization results of residential VPP. Section V concludes the paper.

2 | RESIDENTIAL VPP ANALYSIS

2.1 | 24-h power output prediction

PV output prediction is of great importance for the residential VPP strategy. Literatures [16–18,23] prove that the artificial neural networks are precise methods to predict PV output power. In this work, We are mainly concerned about the hourly energy output prediction in the next day of a residential PV system.

2.1.1 | Weather type classification

The different weather types can lead to significant change in PV system output power. Figure 1 displays the hourly

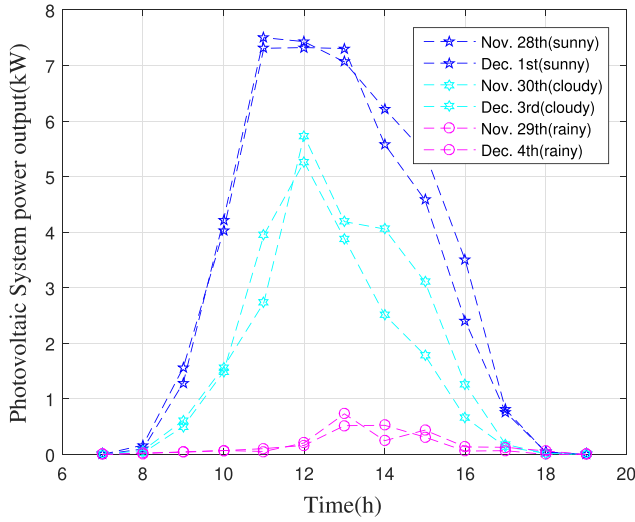


FIGURE 1 Hourly photovoltaic output power with three different weather types [Color figure can be viewed at wileyonlinelibrary.com]

photovoltaic output power for PV system with three different weather types. In order to make the prediction performance of RBF network more effective, a self-organized map (SOM) is applied to classify the PV system output data and numeral weather data into three types. In general, the days of the sunny type have high photovoltaic energy output, and those of the cloudy type have low photovoltaic energy output, while those of the rainy type have lower energy output. Therefore, the overall prediction system contains three models: the sunny photovoltaic power prediction model, the cloudy photovoltaic power prediction model, and the rainy photovoltaic power prediction model. These models are described in detail below.

2.1.2 | RBF network forecasting model

Figure 2 shows the radical basis function (RBF) network with m neurons in input layer, l neurons in hidden layer, and n neurons in output layer. Let $x : [x_1, x_2, \dots, x_m]$, $h : [h_1, h_2, \dots, h_l]$ and $y : [y_1, y_2, \dots, y_n]$ are the input vector, the output of l units in the hidden layer, and the output of the network. According to the input pattern of the RBF network [18], $[h_1, h_2, \dots, h_l]$ of the output in the hidden layer equals to:

$$H_i(x) = \exp\left(-\frac{1}{\sigma_i^2} \|x - \hat{x}_i\|\right), i = 1, 2, \dots, l, \quad (1)$$

where $\hat{x}_i$ is the center of i th RBF unit; σ_i is the width of i th RBF unit. The output value y is the linear combination of the weighted outputs of the hidden units, and is equal to:

$$y_j = \sum_{i=1}^l \omega_{ij} H_i(x), j = 1, 2, \dots, n, \quad (2)$$

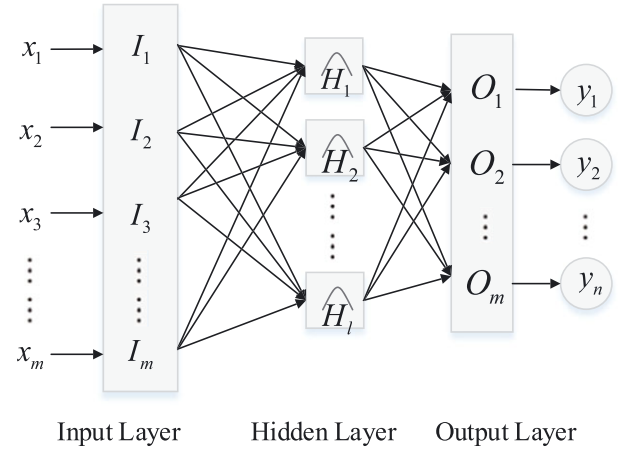


FIGURE 2 The model of RBF network [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 The correlation coefficient between the meteorological factors and the daily energy of a PV system

Correlation	Daily energy
Daily Energy	1
Daily Irradiation	0.9465
Atmospheric Pressure	-0.1349
Daily Temperature	0.6235
Relative Humidity	-0.6550
Daily Rainfall	-0.0101
Wind Speed	-0.0627

where ω_{ij} is the weight of the connection linking the i th RBF unit and the j th output. In order to improve the prediction accuracy, the input layer should contain the main factors affecting the photovoltaic power output. The data of meteorological factors and the daily energy output of a PV system from August 18th 2006 to November 30th 2006 are analysed by correlation analysis with Pearson measurement methods. The result is shown in Table 1. According to the table, the daily irradiation $I(k)$, daily temperature $T(k)$, relatively humidity $H(k)$ are the main factors affecting the photovoltaic power output. Therefore, the input layer of the RBF network will contain these parameters, which can be obtained by the numerical weather prediction (NWP). The number of the neurons within the hidden layer is optimized and decided through the learning step of the network by setting training parameters mean squared error goal (MSEG). The output layers are the 24-hour photovoltaic power output $P_k(t)$, $t = 1, 2, \dots, 24$. The RBF network structure for predicting hourly photovoltaic power of one day shown in Figure 3 can be expressed as the equation below.

$$(P_k(1), P_k(2), \dots, P_k(24)) = f(I(k), T(k), H(k)) \quad (3)$$

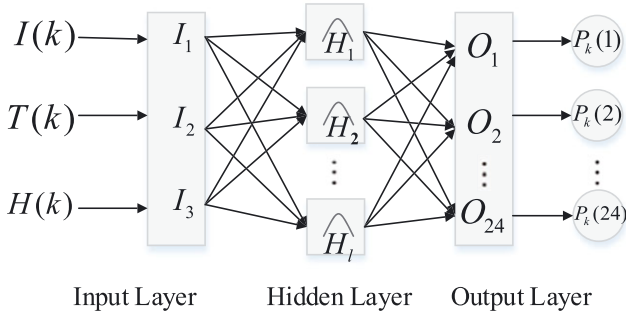


FIGURE 3 The RBF network structure for predicting hourly photovoltaic power [Color figure can be viewed at wileyonlinelibrary.com]

2.2 | Price-based demand response model

Price-based demand response (PBDR) model is utilized to optimize the residents' power consumption behavior and reduce the electricity bills. The model represents the changes of load with respect to change of the price. Load sensitivity with respect to the electricity price is called elasticity which is described as follow [24,25].

$$e_{st} = \frac{\Delta L_s / L_s^0}{\Delta \lambda_t / \lambda_t^0} \begin{cases} e_{st} < 0, & \text{if } s = t \\ e_{st} \geq 0, & \text{if } s \neq t \end{cases} \quad (4)$$

where λ_t^0 and L_s^0 are the initial electricity price at hour t and s , and $\Delta \lambda_t$ is the price change at hour t . When λ_t^0 , L_s^0 , $\Delta \lambda_t$ and e_{st} are known, ΔL_s can be obtained. If the electricity price changes in diverse periods, customers could respond to the alterations in two ways.

1. Some loads (e.g. lighting) are just on or off and cannot be transferred to other periods. This load's elasticity is called self-elasticity and has $s = t$, which always has a negative value [26].
2. Some loads (e.g. washing machine) can be shifted to off-peak periods. This load's elasticity is called cross-elasticity and has $s \neq t$, which always has a positive value [27].

Therefore, the self-elasticity and cross-elasticity can be represented as a square matrix of 24 dimensions as below.

$$\begin{bmatrix} \frac{\Delta L_1}{L_1^0} \\ \frac{\Delta L_2}{L_2^0} \\ \vdots \\ \frac{\Delta L_{24}}{L_{24}^0} \end{bmatrix} = \begin{bmatrix} e_{1,1} & \dots & e_{1,24} \\ \vdots & \ddots & \vdots \\ e_{24,1} & \dots & e_{24,24} \end{bmatrix} \times \begin{bmatrix} \frac{\Delta \lambda_1}{\lambda_1^0} \\ \frac{\Delta \lambda_2}{\lambda_2^0} \\ \vdots \\ \frac{\Delta \lambda_{24}}{\lambda_{24}^0} \end{bmatrix} \quad (5)$$

As a result, the load L_t after PBDR is given as follows.

$$L_t = L_t^0 \times \left\{ 1 + e_{tt} \times \frac{|\lambda_t - \lambda_t^0|}{\lambda_t^0} + \sum_{\substack{s=1 \\ s \neq t}}^{24} e_{st} \times \frac{|\lambda_s - \lambda_s^0|}{\lambda_s^0} \right\}. \quad (6)$$

2.3 | Receding horizon optimization

The approach taken in this work is based on linear programming and receding horizon optimisation [28], which is one of the most widely used techniques in practice. In RHO, an optimal problem is solved at each time step based on estimates available at the current time step. For example, the exact power output of a PV system can be exactly known at the current hour, while the output at next hours can just be estimated. The solution of RHO to the problem provides the optimal sequence of decisions for the finite horizon. Only the first decision is applied, and at next time step the optimisation is repeated based on the current state and the updated predictions for the future states. The Figure 4 shows the basic process of RHO.

3 | RESIDENTIAL VPP MODELING

In this paper, the residential VPP is proposed for optimizing the home energy management aggregating the rooftop PV system, household energy storage devices, and demand response in a TOU electricity market. The goal is to minimize power interaction with the main grid or maximize the profit of trading energy with the market. The main structure of a residential VPP is shown as Figure 5.

The residential VPP has certain capability to be independent of main grid, as the PV system can provide electric energy and the energy storage devices can save electric energy and supply to load temporarily. When the amount of this kind of resident consumers becomes large, peak load shifting can be achieved. The objective of minimizing

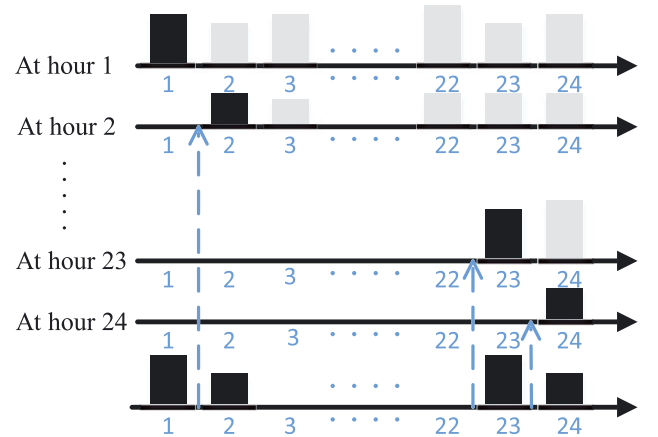


FIGURE 4 The basic process of receding horizon optimisation [Color figure can be viewed at wileyonlinelibrary.com]

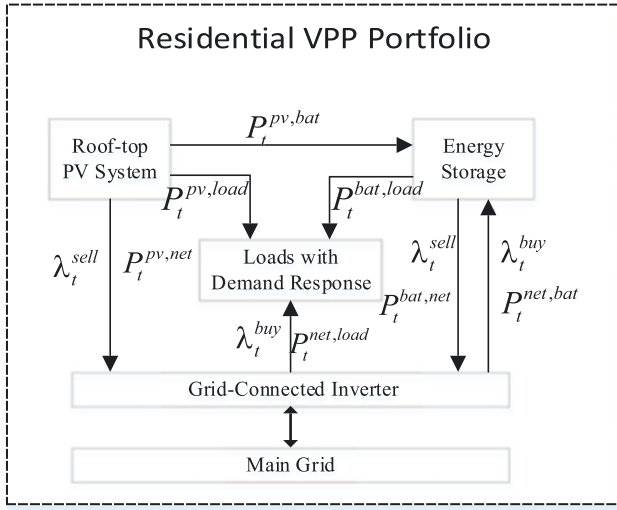


FIGURE 5 The portfolio of residential VPP [Color figure can be viewed at wileyonlinelibrary.com]

the power interactive with the main grid [3,10] is formulated as follows:

$$\min G_1 = \sum_{t=1}^{24} (P_t^{pv,net} + P_t^{net,load} + P_t^{bat,net} + P_t^{net,bat}) \quad (7)$$

The residential VPP can play an important role in peak load shifting. In one day, the consumers can save surplus energy of PV system and absorb energy from main grid at periods of low prices and discharge to main grid at periods of high prices. This incentive brings the objective of maximizing the profit by selling surplus energy.

$$\max G_2 = \sum_{t=1}^{24} (\lambda_t^{sell} P_t^{pv,net} - \lambda_t^{buy} P_t^{net,load} + \lambda_t^{sell} P_t^{bat,net} - \lambda_t^{buy} P_t^{net,bat}) \quad (8)$$

Two objective functions can be formulated in one equation by adding a weight parameter $\delta \in [0, 1]$.

$$\min G = \min (\Delta G_1 + (1 - \Delta)(-G_2)) \quad (9)$$

The bigger weight parameter δ means the higher importance of goal G_1 in (7), thus the minimization operation will definitely tends to optimize G_1 which aims to reduce the energy interactive between the household VPP and main grid. Similarly, the smaller weight parameter δ means the higher importance of goal G_2 in (8), thus the minimization operation will increase the economical profit as much as possible. The constraints for the residential VPP include the supply and demand balance constraint, the capacity constraint of energy storage, the

constraints of charging and discharging limits for energy storage.

1. Supply and demand balance constraint

$$P_t^{pv,load} + P_t^{bat,load} + P_t^{net,load} = L_t, \quad (10)$$

2. Energy storage constraints

The energy storage devices (e.g. batteries) have capacity limits with the consideration of charging and discharging efficiencies. Stored electrical energy at hour t and its limits meet the following constraints:

$$S_t = S_{t-1} + \eta_{chr} P_t^{pv,bat} + \eta_{chr} P_t^{net,bat} - \frac{1}{\eta_{dis}} P_t^{bat,load} - \frac{1}{\eta_{dis}} P_t^{bat,net} \quad (11)$$

$$S_t \geq \bar{S}_{min} \quad (12)$$

$$S_t \leq \bar{S}_{min} \quad (13)$$

3. Energy storage charge and discharge limit constraints

To protect the energy storage devices' lifetime (e.g. batteries), the charging and discharging cannot exceed the maximum capacity as follows.

$$S_t - S_{t-1} \leq \Delta_{chr} \quad (14)$$

$$S_{t-1} - S_t \leq \Delta_{dis} \quad (15)$$

The optimization problem can be modeled as a linear programming problem, including linear objection functions (7)(8)(9) and linear constraints (12)(13)(14)(15). Considering the forecasting error of PV system and hourly residential load, the optimization problem has to be calculated with linear programming for the rest hours based on the updated information of PV system and energy storage. Therefore, the algorithm combining the linear programming and receding horizon optimization is designed for 24-hour scheduling of residential VPP. The complete optimization process is shown in Figure 6. Linear programming provides a reference scheme of $[t, 24]$, and practical operation has to adjust the scheme according to the latest forecasting error and status of energy storage and electricity prices. Energy output of PV system must be firstly used for self-consumption of residential VPP. Charging should be a priority for energy storage (e.g. batteries). The discharging quantity should take the smaller one between its capacity and the reference scheme. Keeping the balance of supply and demand is the most important.

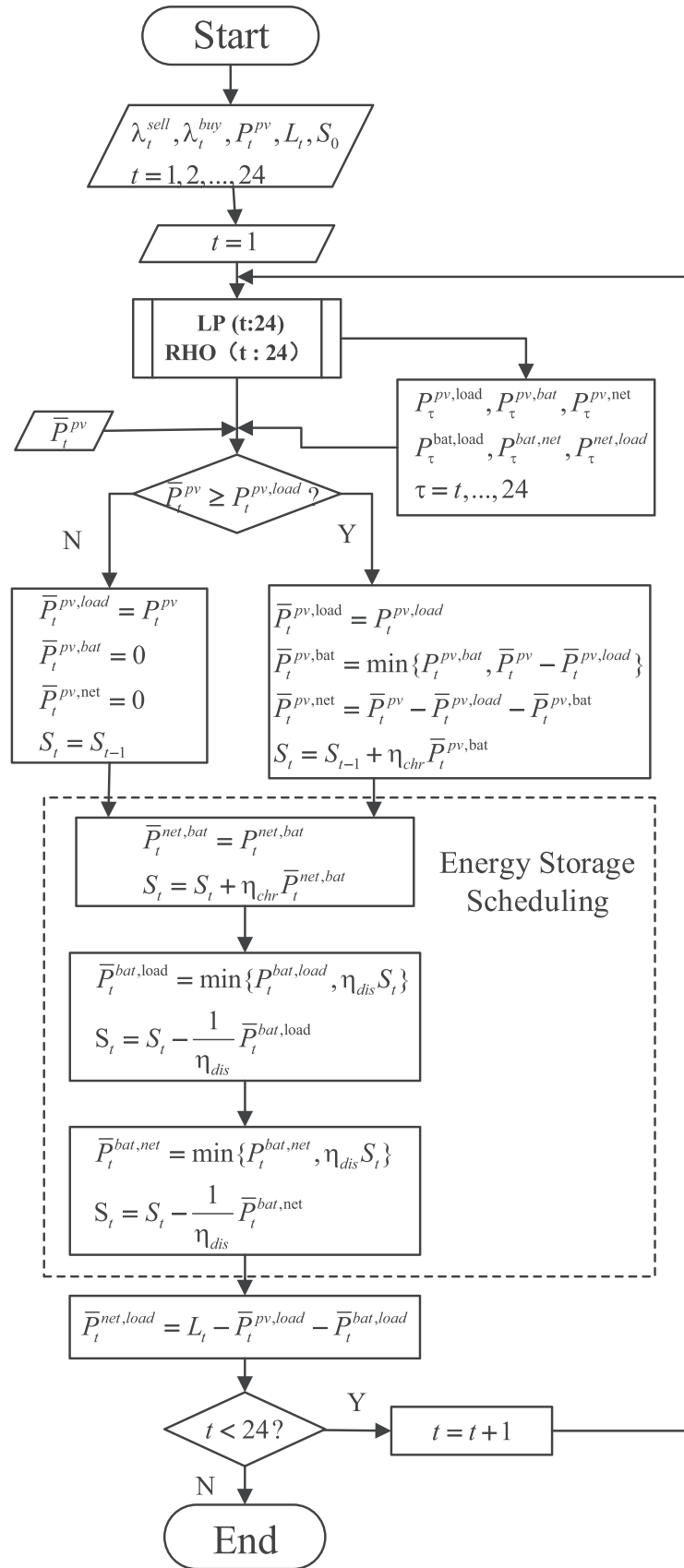


FIGURE 6 The flowchart of RHO with LP for residential VPP

TABLE 2 Demand price elasticity and TOU electricity prices

	Valley	Off-peak	Peak	Prices
Valley	−0.1	0.01	0.012	0.278
Off-peak	0.01	−0.1	0.016	0.58
Peak	0.012	0.016	−0.1	1.044
Periods	1-8	9-18	19-22	
		23-24		

4 | SIMULATION ANALYSIS

4.1 | Basic data

The PV system is located in the Renewable Energy Research Center (RERC) of Huazhong University of Science and Technology (HUST). The installed capacity of photovoltaic system is around 15kW, and data collected in the PV system was recorded including every 5 min power delivered by the PV power system to the grid. The climate data from historical record and online meteorological services includes solar irradiance, air temperature, relatively humidity, wind speed, sunshine duration, air pressure, etc. Demand price elasticity parameters have been extracted from [24,25], and are shown in Table. 2, which also includes a realistic TOU electricity policy where the price in peak periods is 1.8 times of that in off-peak periods, while price in valley is 0.48 times in Shanghai 2009. The price selling power to grid is usually lower than buying power. The motivation is to encourage self-consumption of PV energy. Residential electricity demands are selected from residential service center [29]. Powerwall 2 (called battery later) of Tesla [30] is selected as residential energy storage. Its usable capacity is 13.5 kWh, and the battery has a 90% efficiency of charge and discharge with 100% depth of discharge. The proposed LP-based algorithm is a global optimization method, and runs in a Matlab 2015b environment with embedded LP function.

4.2 | 24-hour PV output forecasting

In order to evaluate the forecasted result, two parameters for the 24-h forecasting can be calculated: the correlation coefficient and the mean absolute percentage error (MAPE). The correlation coefficient is used to illustrate the correlation between variables. The closer the parameter approaches 1, the stronger the correlation is between the forecasted PV power output and the actual output. MAPE provides information on the short performance and represents a measure of the variation between forecasted PV power output and the recorded data. Table 3 shows the the correlation coefficients and MAPE of the forecasted power output and the recorded data for 4 sunny days (Dec. 19th–22nd 2006), 4 cloudy days (Dec. 23th–2th 2006) and 4 rainy days (Jan. 11st–14th 2007). According to the table,

TABLE 3 Correlation coefficient and mean absolute percentage error between forecasted power output and recorded data

	Weather type	Correlation coefficient(%)	MAPE(%)
1	Sunny	96.56	5.30
2	Sunny	99.21	2.30
3	Sunny	99.27	3.27
4	Sunny	96.92	8.80
5	Cloudy	96.35	9.13
6	Cloudy	99.46	2.68
7	Cloudy	98.44	6.37
8	Cloudy	96.79	5.30
9	Rainy	92.76	11.13
10	Rainy	89.15	10.60
11	Rainy	97.08	8.14
12	Rainy	81.21	5.52

the correlation coefficient varies in [96% – 99%] and MAPE varies in [2% – 10%] in sunny and cloudy days. This shows the RBF network for forecasting the PV power output is very suitable for sunny and cloudy days, while it is also acceptable for rainy days. Figure 7A, Figure 7B and Figure 7C shows the forecasting results for the mentioned sunny, cloudy and rainy days.

4.3 | Demand response

The PBDR described in III is usually utilized to optimize the residents' power consumption behavior according to the TOU electricity price. Figure 8A shows the forecasted daily demand and the daily demand after PBDR for a typical day of August. Figure 8B shows the forecasted daily demand and the daily demand after PBDR for a typical day of January. As can be seen, the demand curves after PBDR in peak hours are slightly lower than those forecasted curves, and the reduced demand has been transferred to other periods. This is in the interests of consumers they use less electricity power either from the main grid or the battery in peak hours, in which the price is high. Moreover, they can sell more energy stored in the battery to main grid for economy, and in this way it can reduce the demand burden of main grid.

4.4 | Residential VPP scheduling

The residential VPP optimizes the home energy management aggregating the rooftop PV system, residential energy storage device (battery) and considering demand response in a TOU electricity market. The goal is to minimize the power purchased from main grid or maximize the profit by selling the renewable energy to main grid (the same as minimize the electricity consumption cost). According to the previous analysis, several typical types of scenarios, e.g. summer sunny day (S.S.), sum-

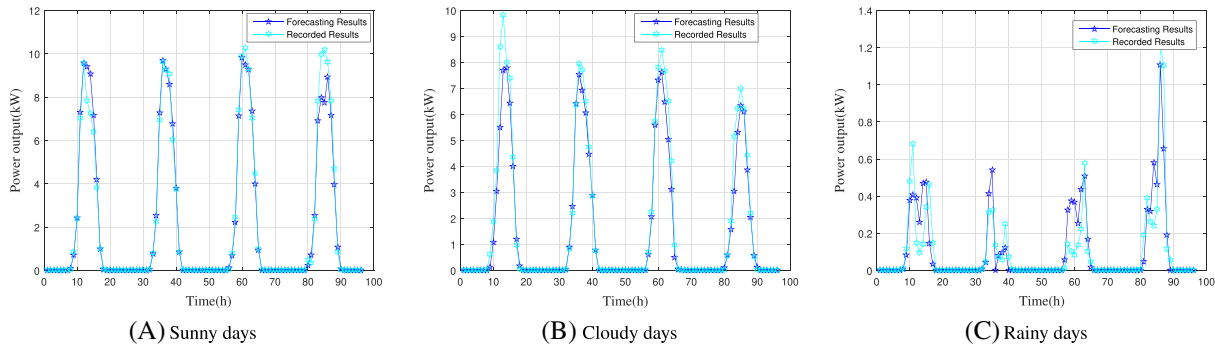


FIGURE 7 Forecasting results of the RBF network compared with recorded data for Sunny, Cloudy, and Rainy days [Color figure can be viewed at wileyonlinelibrary.com]

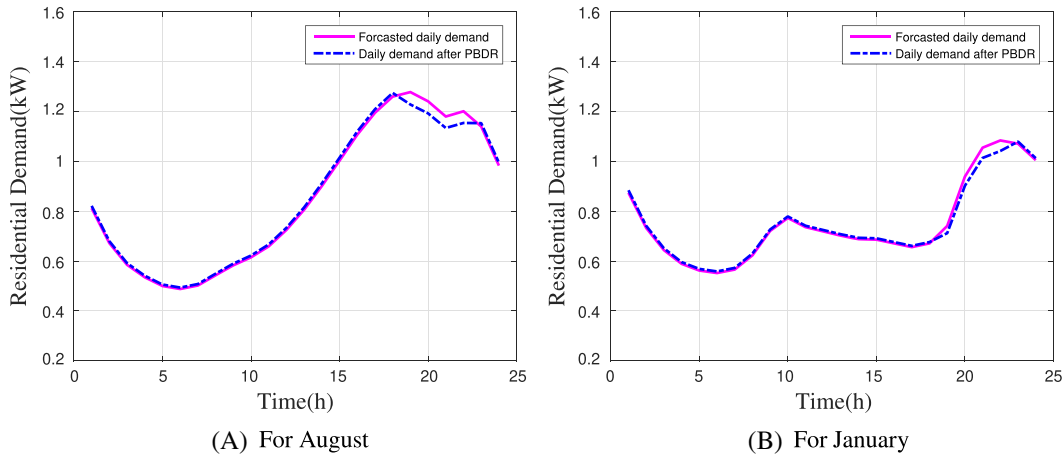


FIGURE 8 Forecasted daily demand and daily demand after PBDR for August and January [Color figure can be viewed at wileyonlinelibrary.com]

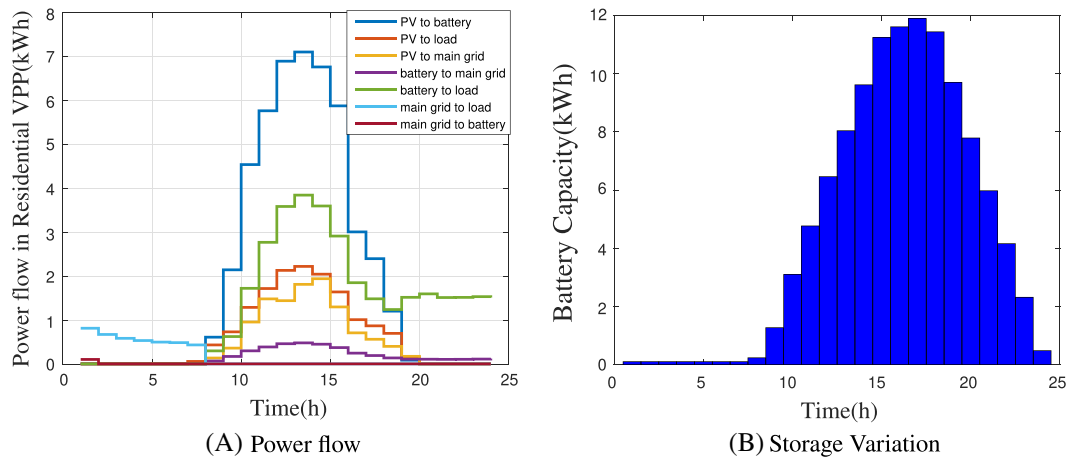


FIGURE 9 Electric power flow and storage variations in 24 hours for typical summer sunny day with $\delta = 1$ [Color figure can be viewed at wileyonlinelibrary.com]

mer cloudy day (S.C.), summer rainy day (S.R.), winter sunny day (W.S.), winter cloudy day (W.C.) and winter rainy day (W.R.), should be considered separately, while other factors, e.g. demand response and parameter δ in Equation (9) also require discussion.

4.4.1 | Detailed analysis for one typical scenario

A typical scenario of summer sunny day is selected to discuss the influence of demand response and parameter δ .

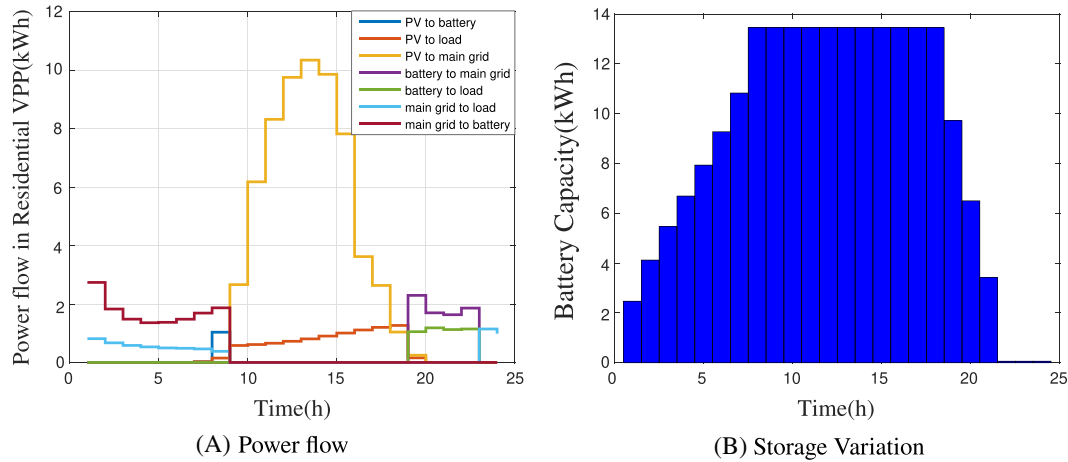


FIGURE 10 Electric power flow and storage variations in 24 hours for typical summer sunny day with $\delta = 0$ [Color figure can be viewed at wileyonlinelibrary.com]

Figure 9A, Figure 9B show the electric power flow details and storage variations in residential VPP for 24 hours with goal of minimizing power purchased from main grid ($\delta = 1$), while Figure 10A, Figure 10B are those with the economical goal of the profit by selling energy to main grid $\delta = 0$. Firstly, the lines of energy from PV to man grid, from battery to main grid, from main grid to load and from main grid to battery in Figure 9A are obviously lower than those in Figure 10A. Secondly, the battery is mainly charged with the local PV energy and discharged for local utilization when $\delta = 1$ while the battery is charged both from PV and main grid and also partly discharged to main grid for arbitrage when $\delta = 1$. These results show that when $\delta = 1$ VPP tends to reduce the energy interactive with main grid to reduce reliance while when $\delta = 1$ VPP tends to maximize its profit through energy interactive with main grid in a TOU energy market. The conclusion illustrates the contradiction of the presented two goals.

Table 4 shows the profits by selling power and purchased power from the main grid with and without DR in different $\delta \in [0, 1]$. Figure 11 shows the comparison in a bar graph. It can be seen that $\delta \in [0.3, 0.9]$ with DR is the best choice. In this situation, one part of surplus PV output power is sold to main grid, and another part is charged to battery for home demand at peak hours. According to Table 4, PBDR indeed brings some benefits of profit, which is reasonable that PBDR reduce the demand in peak hours. It can be summarized that residential VPP scheme provided by the proposed algorithm is applicable with high profit and less purchasing power from main grid when $\delta \in [0.3, 0.9]$ in a typical summer sunny day which usually has high daily PV power output and high daily demand.

TABLE 4 Profits and purchased power with and without DR with different δ

δ	Profit (¥)		Purchased power (kWh)	
	With DR	Without DR	With DR	Without DR
1	6.334	6.343	4.187	4.139
0.3-0.9	26.663	26.574	4.187	4.139
0.2	27.428	27.330	6.335	6.262
0.1	30.220	30.121	20.527	20.448
0	30.220	30.121	20.527	20.448

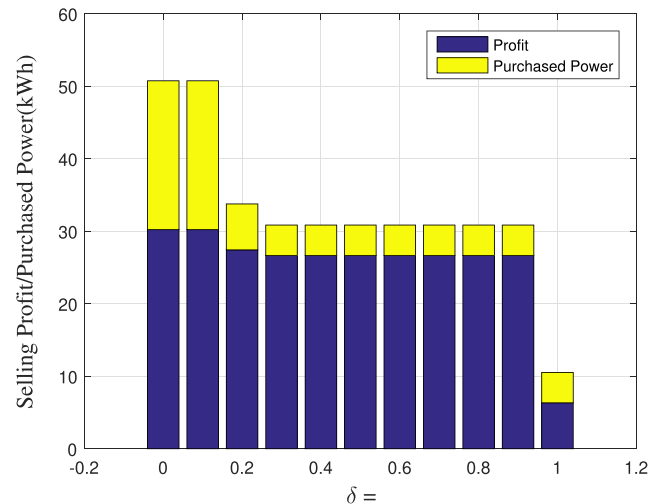


FIGURE 11 Profits and Purchased Power Comparison with DR with different δ [Color figure can be viewed at wileyonlinelibrary.com]

4.4.2 | Comparison for different types of scenarios

Six typical scenarios for residential VPP are simulated and discussed in detail. As Table 5 shows, summer sunny days usually have most output power of PV system, which not only provides for residential demand but also brings high

TABLE 5 Profits and purchased power in six typical scenarios with different δ

δ Type	$\delta = 1$		$\delta = 0.5$		$\delta = 0$	
	Profits	Power	Profits	Power	Profits	Power
S.S.	6.334	4.187	26.663	4.187	30.220	20.267
S.C.	5.219	4.097	19.378	4.097	22.867	20.213
S.R.	3.985	4.031	16.741	4.031	20.297	20.851
W.S.	4.325	5.608	18.796	5.616	22.672	23.178
W.C.	2.180	5.285	10.557	5.318	14.389	22.71
W.R.	-7.94	15.13	-5.44	15.98	-0.49	26.43

profit. Without high output power, other scenarios have less profit and have to purchase more energy for demand. In these scenarios, residential VPP can reduce the cost of purchasing the power by scheduling the storage and providing its energy to main grid in peak hours. Winter rainy days have little PV output, and have to pay for the power energy provided by main grid. In most of the scenarios, residential VPP with rooftop PV system and energy storage can purchase less power from main grid and gain high profit by selling surplus renewable energy. Besides reducing dependence on grid, it is also helpful for relieving the burden of main grid in peak hours.

5 | CONCLUSIONS

In this paper, we propose a residential VPP to optimize the residential energy management considering rooftop PV system, energy storage and demand response. The algorithm combining receding horizon optimization and linear programming provides hourly optimal energy management scheme. The self-organizing map and radical basis function are adopted to classify the weather type and predict hourly power output of PV system more precisely. In a time-of-use electricity market, price-based demand response is analysed to adjust the resident's demand. Modeling of residential VPP have to satisfy the constraints of balance of supply and demand, battery capacity limits and charging and discharging limits. By setting a parameter to control the optimization objective, an optimal goal can be achieved with the parameter equal 0.3 – 0.9. Simulation cases have been carried out to verify the practicability of residential VPP and the algorithm. The energy schedule of multiple houses will be our research focus in future work. We will also pay attention to the technical and economic feasibility analysis of battery energy system, and the technical feasibility of the application of DR in a long-term energy schedule in future work.

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ORCID

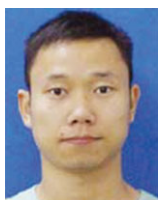
Yan-Wu Wang  <https://orcid.org/0000-0002-4722-1332>

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AUTHOR BIOGRAPHIES



Shichang Cui received the B.S. degree in Automation from the Huazhong University of Science and Technology, Wuhan, China, in 2016, where he is currently pursuing the Ph.D. degree in control science and engineering. His current research interests include robust

optimization, stochastic optimization, and energy management for smart grids.



Yan-Wu Wang (M_i10 – SM_i13) received the B.S. degree in automatic control, the M.S. degree and the Ph.D. degree in control theory and control engineering from Huazhong University of Science and Technology (HUST), Wuhan, China, in 1997, 2000, and 2003, respectively. Currently, she is a Professor with the School of Automation, HUST, and with the Key Laboratory of Image Processing and Intelligent Control, Ministry of Education, China. Her research interests include hybrid systems, cooperative control, and multi-agent systems, with applications in smart grid.

Dr. Wang was a recipient of several awards, including the first prize of Hubei Province Natural Science in 2014, the first prize of the Ministry of Education of China in 2005, and the Excellent PhD Dissertation of Hubei Province in 2004, China. In 2008, she was awarded the title of "New Century Excellent Talents" by the Chinese Ministry of Education.



Xiangning Lin was born in Guangxi, PR. China, in 1970. He received the M.Sc. and Ph.D. degrees in electrical engineering from HUST. Currently, he is a Professor at HUST. His research interests are modern signal processing and power system protective relaying and microgrid.



Jiang-Wen Xiao received the B.S. degree in electrical engineering, the M.S. degree in control theory and control engineering, and the Ph.D. degree in systems engineering from Huazhong University of Science and Technology (HUST), Wuhan, China, in 1994, 1997, and 2001 respectively. From 2001 to 2003, he worked as a Post-Doctoral Researcher in the College of Management, HUST, Wuhan, China. He is currently a Professor in the School of Automation, HUST. His research interests include modeling and control of nonlinear and complex systems, data mining, and optimization.

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